

Azure +FSD

Duration 22 Days

Objectives:

- Candidates are responsible for participating in all phases of development including requirements gathering, design, development, deployment, security, maintenance, performance tuning, and monitoring.
- They should be proficient in Azure: SDKs, data storage options, data connections, APIs, app authentication/authorization, computer/container deployment, and debugging. Candidates partner with cloud solution architects, DBAs, DevOps, infrastructure admins, and other stakeholders to implement solutions.
- Candidates should have at least 2 years of professional development experience and experience with Microsoft Azure. They should be able to program in an Azure-supported language, and should be proficient using Azure CLI, Azure PowerShell, and other tools.
- Gain a comprehensive understanding of the Angular framework and its key concepts.
- Learn to set up Angular development environments and use Angular CLI effectively.
- Master Angular architecture, components, services, forms, routing, and HTTP client.
- Explore advanced topics such as state management with NgRx, authentication, optimization, testing, and deployment.
- Build practical skills through hands-on exercises and real-world projects.

Prerequisites:

- Familiarity with HTML, CSS, and JavaScript fundamentals.
- Basic understanding of TypeScript will be beneficial but not mandatory.
- Prior experience with any programming language is advantageous.
- Comfortable using a code editor and command-line interface.

Outcomes:

- Understand the core concepts and architecture of Angular.
- Be proficient in setting up Angular development environments and creating Angular applications using Angular CLI.
- Develop skills in building components, forms, and services, and managing data flow within Angular applications.
- Implement routing and navigation, handle HTTP requests, and integrate external APIs.
- Gain expertise in advanced Angular topics including state management with NgRx, authentication, optimization techniques, testing methodologies, and deployment strategies.
- Acquire practical experience through hands-on exercises and projects, enabling them to develop robust and scalable web applications with Angular.

Day 1

Getting Started with Angular

- Introduction to Angular
- Overview of Angular framework.
- History and evolution of Angular.
- Setting Up Angular Environment
- Installing Node.js and npm.
- Installing Angular CLI.
- Creating your first Angular application.
- Angular Architecture Overview
- Understanding Angular modules and components.
- Introduction to TypeScript basics.
- Components and Templates
- Creating components.
- Using templates and data binding.
- Interpolation and property binding.

Day 2

- RxJS operators for data manipulation
- Directives and Pipes
- Understanding built-in directives (ngIf, ngFor, etc.).
- Creating custom directives.
- Using pipes for data transformation.

Day 3

- Advanced Angular Concepts
- Forms in Angular
- Template-driven vs. Reactive forms.
- Form validation and error handling.
- Working with form controls and data binding.

Day 4

- Services and Dependency Injection
- Creating Angular services.
- Injecting services using Dependency Injection (DI).
- Sharing data between components using services.
- Routing and Navigation
- Setting up routing in Angular.
- Configuring route parameters and nested routes.
- **Configuring route guards and access control.**
- Implementing navigation guards.

- HTTP Client in Angular
- Making HTTP requests with HttpClient.
- Handling responses and error handling.
- Using Observables for asynchronous operations.

Day 5

Building Angular Applications

- Angular CLI and Project Structure
- Using Angular CLI commands for project management.
- Understanding Angular project structure and files.
- Component Communication
- Parent-child component communication using @Input and @Output.
- Communication between unrelated components using a service.
- State Management with RxJS operators
- Introduction to NgRx for state management.
- Setting up NgRx store, actions, reducers, and effects.
- Managing application state with NgRx.
- Advanced Angular Topics
- Authentication and Authorization
- Implementing authentication with Angular (JWT, OAuth).
- Protecting routes with guards.
- Role-based access control (RBAC) in Angular applications.
- Angular Best Practices and Optimization
- Performance optimization techniques.
- Code organization and maintainability.
- Handling internationalization (i18n) and localization.

Day 6

Advanced Git

- Limitations of existing Version Control Systems
- Installing and Configuring Git
- Git Configuration Levels
- Creating a Git Repository
- Git Workflow
- Tracking File Changes
- Reverting to Earlier Commits
- Deleting Files in Git
- Ignoring Files in Git
- Renaming Files in Git
- Branching in Git

- Merging Branches in Git
- Fast Forward and Recursive Merge
- Recursive Merge
- Resolving Merge Conflicts in Git
- Stashing in Git
- Rebasing in Git
- Cloning in Git

Day 7

Python Scripting

- Introduction to Python Scripting for Automation
- System Automation Tasks
- Automating system tasks using Python (os, shutil, subprocess)
- Working with command-line arguments
- Parsing json and extracting information
- Python data structures

Day 8

Implementing Azure Storage Account

- Introduction to Azure Blob, Queue, and Table Storage.
- Setting up storage accounts and managing data.
- Azure Key Vault overview and configuring secrets.
- Integrating Key Vault with applications securely.

Implementing Azure CosmosDB for NOSQL

- Cosmos DB overview and NoSQL basics.
- Setting up a Cosmos DB instance and choosing consistency levels.
- Performing CRUD operations in Cosmos DB.
- Integrating Cosmos DB with applications. Javascript and HTML Pages

Day 9

Implementing Azure Service Bus

- Introduction to Azure Service Bus.
- Creating message queues and topics.
- Configuring Azure Service Bus for message exchange.
- Integrating Service Bus with applications.

Implementing ETL Operations using Azure Data Factory

- Overview of Azure Data Factory (ADF).
- Creating and scheduling data pipelines.
- ETL workflows using ADF.
- Configuring data sources and destinations.

Azure Powershell

- Introduction to Azure PowerShell.
- Writing scripts to automate resource management.
- Using Azure Cloud Shell for quick scripting.
- Automating deployments with PowerShell.

Day 10

Implement Azure App Service Web Apps

- Create an Azure App Service Web App
- Enable diagnostics logging
- Deploy code to a web app
- Configure web app settings including SSL, API settings, and connection strings
- Implement autoscaling

Day 11

Implement Azure Functions

- Create and configure an Azure Function App
- Implement input and output bindings
- Implement function triggers by using data operations, timers, and webhooks

Day 12

Azure CLI

- Fundamental concepts, installation and configuration of Azure CLI, PS, Bash
- Basic Azure CLI, PS, Bash commands (login, account, group, resource)
- Configuring authentication methods (Azure AD, service principal, managed identity) using Azure CLI
- Understanding Azure Resource Manager (ARM) for Cloud Native development
- Creating and managing resource groups and resource types
- Creating, updating, and deleting resources using Azure CLI
- Working with resource properties and tags with Azure CLI
- Using Azure CLI to query Azure resources

Day 13

YAML Scripting

- Reading and writing YAML in Python.
- Loading and dumping YAML using pyyaml
- Using YAML for configuration files (e.g., Docker, Kubernetes)
- YAML for defining CI/CD pipelines (e.g., GitHub Actions, Azure Pipelines)
- Best practices for writing YAML files in DevOps

Day 14

Azure DevOps

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- Introduction to DevOps life cycle and Azure DevOps
- Connecting Azure CLI to Azure DevOps
- Azure Project management dashboards – Kanban, Agile, Scrum, CMMI
- User Stories – SCRUM, Agile, CMMI
- Epic, Issue, Bugs, Tasks, Subtasks,
- Plugins, Extensions, Building Kusto queries

Day 15

Azure Repos

- Introduction and configuration of Azure repos
- Configuring Source Control, Version Control and Service connections
- Configuring Repo connectivity
- Configuring Git, GitHub, Bitbucket, SVN,

Day 16

Azure Build

- Introduction to Azure Pipelines
- Creating build pipelines (Classic and YAML)
- Continuous Integration (CI) with Azure Build
- Integrating with GitHub, Azure Repos
- Pipeline triggers (scheduled, pull requests)
- Build agents and agent pools

Day 17

Azure Testplans

- Overview of Azure Test Plans and its role in DevOps
- Manual, automated, and exploratory testing
- Creating and managing test cases, suites, and plans
- Configuring test environments
- Associating tests with work items
- Running manual and automated tests in pipelines
- Integrating automated tests with Azure Pipelines
- Continuous Testing and release gates
- Analyzing test results and tracking bugs
- Load testing and performance testing with Azure
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Day 18

Azure Artifacts

- Introduction to Azure Artifacts
- Package management with Azure Artifacts (NuGet, npm, Maven, Python)
- Creating, publishing, and consuming packages
- Versioning and dependency management
- Integrating Artifacts with build pipelines
- Retention policies and cleanup strategies
- Controlling access to artifacts (scoped feeds, permissions)
- Setting up upstream sources
- Best practices for artifact management in large projects
- Introduction to IaC and Azure Resource Manager (ARM).
- Building and deploying resources with ARM templates.
- Overview of Terraform for Azure.
- Best practices for IaC in Azure."

Day 19

Azure Pipelines for CI/CD

- Creating CI Pipeline – Classic and Yaml
- Creating CI Pipeline, defining test and delivery stages
- ADO pipeline for Continuous Deployment/Delivery,
- Working with test results and pipeline
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Day 20

Containerization on Azure

- Container Technology, Container orchestration, Create and Manage containerized services using Docker
- Use existing container images to create and manage containerized services, Docker Architecture and Docker file
- Working with images and Containers, Storage and Persistence Layer, Understanding Dockers
- Docker compose
- Azure Container Instances (ACI)
- Creating and managing container instances
- Networking and security in ACI
- Azure Container Registry (ACR)
- Setting up and managing ACR
- Pushing and pulling container images to/from ACR
- Integrating ACR with ACI

Day 21

Azure Kubernetes Service

- Deploy and manage applications at scale
- Overview of AKS
- Container solutions in Azure
- Features of AKS
- Get started with AKS
- Securely modernize and optimize Kubernetes environments with unified management, governance, and monitoring.
- Build Resilient Global Applications
- Enhance Developer Productivity
- Using Azure Key Vault for secret management in AKS.
- Integrating Key Vault with Kubernetes applications.
- Configuring Key Vault access policies.
- Best practices for securing sensitive data.

Day 22

Testing and Deployment of Angular Applications

- Testing Angular Applications
- Writing unit tests with Jasmine and Karma.
- End-to-end (E2E) testing with Protractor.
- Mocking HTTP requests for testing services.
- Deployment Strategies
- Building Angular applications for production.
- Deploying Angular apps to different platforms (Firebase, AWS, etc.).
- Continuous Integration and Continuous Deployment (CI/CD) pipelines.